From Mass Spectrum to Molecule

Technical brief · October 18, 2026

How the best public model for Kaggle's Enveda CASMI 2026 challenge (Sep 14 – Dec 14 2026, $50k) works, and what our strategy has found so far

In short. The best public pipeline (0.450) *retrieves* known structures of the right mass, scores them with several kinds of spectral evidence, fuses the rankings and breaks isomer ties with a spectrum-predicting model. We reproduced it (A), learned that novel molecules add at most ~0.01 (B), found two train/serve bugs (C) and are building a fully rule-compliant pipeline (D).

1 The problem

A tandem mass spectrometer (LC-MS/MS) ionises a molecule, weighs the intact ion (the **precursor m/z**, mass-to-charge ratio), breaks it by gas collisions (20, 40 or 60 eV) and weighs the charged pieces, the **fragments**. A **spectrum** is the list of (fragment m/z, intensity) peaks. Each hidden test molecule (natural-product-like) has 1–16 spectra (median 3, Bruker timsTOF). **Task:** return up to 25 ranked structure guesses (SMILES).

Score, MRR@25 (mean reciprocal rank): the average over molecules of 1/rank of the first correct guess (rank 2 → 0.5; not in top 25 → 0). "Correct" = same atom connectivity (first InChIKey block; stereochemistry ignored).

Three test classes (shares hidden; community estimates):

- **Class 1** (~16–28%): public reference spectra exist → library look-up works (MRR ~0.93).
- **Class 2** (~20–45%): in PubChem or COCONUT but never measured → retrieve, then rank above same-mass rivals.
- **Class 3** (~39–52%): new, not in PubChem → must be *generated*.

Data and rules. 2.54M training spectra of ~275,810 structures from 11 public libraries. Code runs offline in a Kaggle notebook (≤ 9 h); for prizes, data and weights must be public, free, commercially usable and reproducible (no NIST).

A noisy scoreboard. ~400 hidden molecules; the public board uses only ~124 (31%), so one molecule going from miss to rank 1 is worth ~0.008 (standard error ~0.016). Oct 17: top 0.495, best public notebooks 0.44–0.45, 3,486 teams.

KEY TERMS

Adduct: the charged form measured, e.g. $[M+H]^+$ (+ proton) or $[M+Na]^+$; 10 allowed. It turns precursor m/z into neutral mass M.
Isomers: different structures with one formula, hence one mass.
Fingerprint: one bit per substructure (present/absent). *Tanimoto* = share of bits in common.
Analog: training molecule whose spectrum looks similar.
Forward model: predicts the spectrum of a given structure.
ppm: parts per million of the mass (10 ppm of 400 Da = 0.004 Da).

2 How the current public model works

The strongest public notebook ("From Spectra to Structures" v51, 0.450, "E006" lineage) stacks many community contributions (Fig. 1). It outputs structures from its databases, plus a few from a small generator.

1. **Mass → candidates.** M = median neutral mass over the spectra; candidates = all structures of a ~711k pool (training + COCONUT natural products) within 10 ppm of M (30 ppm fallback): tens to hundreds, mostly isomers (Fig. 2).
2. **Evidence (Engine 1)**, ~100 features per candidate:
   - *library*: spectral-entropy similarity to the candidate's own reference spectra (solves class 1);
   - *analog propagation*: similar training spectra (after a mass shift) vote for candidates resembling their molecules;
   - *FPNet*: a transformer over peaks + metadata predicts the fingerprint; candidates that agree score high;
   - *fragmentation*: share of peak intensity explained by breaking 1–2 bonds, plus a learned fragment scorer;
   - a class-3 generator (edits of the 6 nearest analogs).

   Four LightGBM gradient-boosted models combine them, then z(score) + 0.15 × popularity (log PubChem substance + literature counts); top 60 kept.
3. **PubChem channel.** Fingerprint models screen ~105M PubChem structures (z + 0.25 × popularity); picks go to ranks 2, 4, 6, 8, 10, to rank 1 if the margin is > 150 or well documented, and are skipped if a library match ≥ 0.9 exists.
4. **Engine 2.** Own pool (adds ChEBI, LIPID MAPS) and channels; two boosted rankers blended 0.88/0.12; top 40.
5. **Fusion.** Reciprocal-rank fusion (each list adds 1/(constant + rank)), duplicates merged. Inside each formula group, forward models ICEBERG/GLACIER predict each candidate's spectrum; the best match to the measured one moves up. Engine 2's top pick goes to rank 2.

History: 0.34 (mid Sep) → 0.40 (engine fusion) → 0.44 (forward-model re-ranking) → 0.45 (tuned PubChem promotion).

Weak spots: almost no class-3 hits; many misses are the *wrong isomer*; tweaks were accepted on the noisy 124-molecule board without cross-validation; some bundled weights have uncertain licence/reproducibility (e.g. CC-BY-NC-SA).

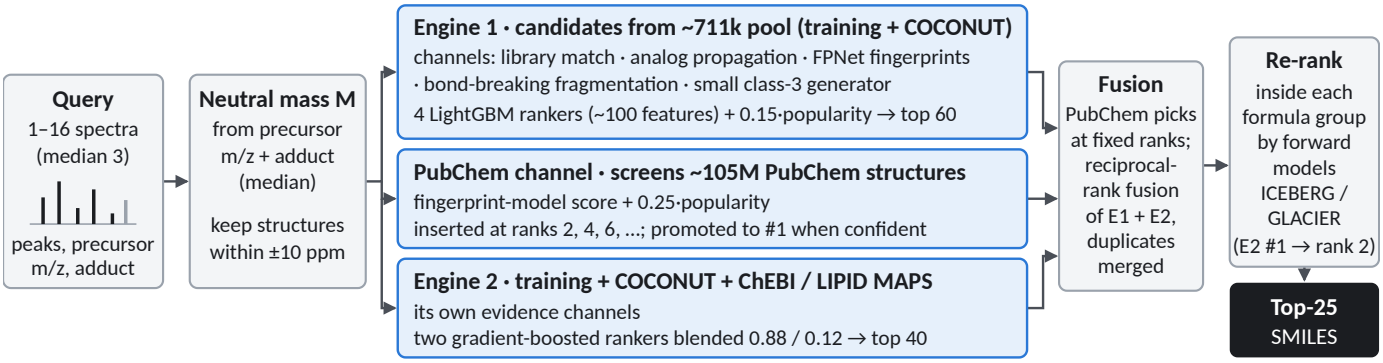


Figure 1. The public model as a pipeline. Blue boxes generate and rank candidates independently and decide *which* known structures enter the list; the last step only reorders candidates sharing a formula (isomers), where mass cannot help.

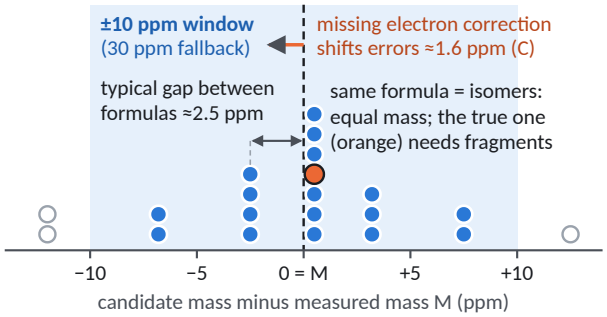


Figure 2. Mass window and candidates (schematic). Dots are database structures, columns are formulas. The mass filter leaves stacks of isomers, so ranking *inside* a stack decides the score.

3 Our strategy (first session, Oct 17–18)

A Get on the board. Forked the 0.450 notebook unchanged, keeping its pinned Docker image (teams running copies on a newer image lost ~0.04). Score pending.

B Class-3 deep-dive (largest untapped share).

- **Proxy:** 16,917 training structures are absent from PubChem's 106M connectivity keys: real measured molecules, the closest stand-in for class 3. 45% split at one amide/ester bond into two PubChem-known pieces (N-acyl amino acids, bile-acid conjugates, fatty-acid esters).
- **NovelEngine** outputs only structures absent from PubChem, COCONUT and training. (i) *Analog derivatives*: similar training spectra, found by modified cosine (which also pairs peaks shifted by the mass difference Δm), are edited by transformations (hydroxylation, methylation, glycosylation, …) whose mass change equals Δm; top-20 analogs reach 24% of the proxy. (ii) *Conjugates*: known acid + known amine/alcohol whose masses add up, ranked by each half's diagnostic fragments.
- **Site localisation** (our new scoring idea, Fig. 3): derivative MRR on the proxy 0.061 → 0.107. A learned LambdaRank ranker: proxy MRR 0.124 → 0.269.
- **Reality check:** a novel-only submission scored 0.010 public vs ~0.045 predicted: the proxy overstates reach ~4–5×; class 3 is worth at most ~0.01 as an add-on (known-only run of the same engine: 0.085).

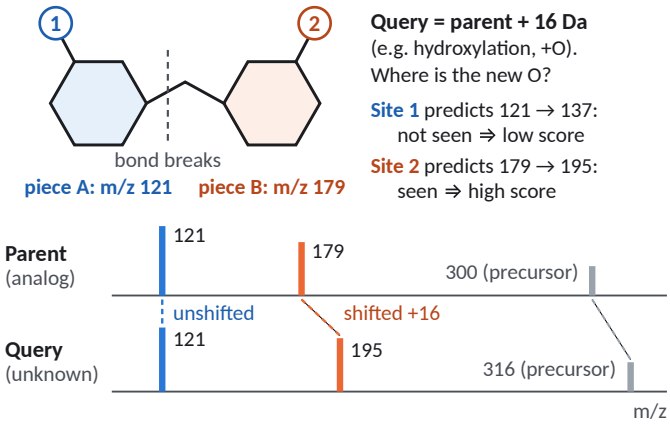


Figure 3. Site localisation (toy numbers). Parent peaks that reappear unshifted come from pieces without the modification, peaks shifted by Δm from pieces carrying it; each candidate site is scored by how consistent all peaks are with it.

C Bug hunt on the public pipeline (subagent code review) found two train/serve mismatches. (1) The ranker feature "number of peaks" uses raw test peaks (median 230) but was trained on cleaned peaks (median 52). (2) Models were fitted on enveda-180 masses *without* the electron-mass correction

D Compliance-first pipeline of our own (all code and weights ours, reproducible, permitted data).

- **Pool** of 726,198 structures (training + COCONUT Oct-2026). **Channels:** library entropy similarity, analog propagation, fragment explanation, COCONUT documentation priors (organisms, literature, synonyms), and our own model.
- **Dual encoder** (Fig. 4), trained to rank the true structure above up to 64 same-mass "hard negatives" (contrastive loss) and to predict fingerprint bits. After one CPU epoch: **MRR 0.51 (top-1 36%)** on held-out structures among their same-mass candidates; longer CPU/GPU runs in progress.
- **Ranker:** LambdaRank (LightGBM trained to order lists) on simulated queries from 8,000 training structures (class 1: hide the query spectra; class 2: hide all spectra of the structure); validated on 1,500 disjoint structures (250 enveda-np-examples, 750 natural products, 500 others). Uses the *measured* mass (the true one would leak the answer) and the electron correction.

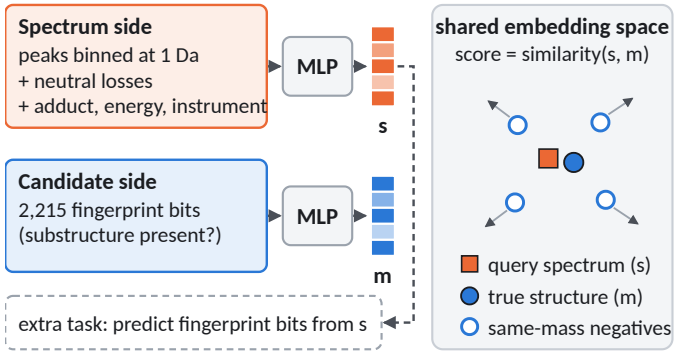


Figure 4. Our dual-encoder model. Two small networks map a spectrum and a candidate structure to vectors s and m. Training pulls true pairs together and pushes same-mass decoys apart; at test time, candidates in the mass window are ranked by similarity to the query.

4 Results so far & lessons

	Experiment (measured on)	Result
A	Fork of 0.450 notebook (board)	*pending*
B	Analog-derivative reach (proxy)	24%
B	Site localisation (proxy MRR)	0.061 → 0.107
B	Learned ranker (proxy MRR)	0.124 → 0.269
B	Novel-only submission (board)	0.010 (~0.045 pred.)
B	Known-only, same engine (board)	0.085
C	Two bug-patched variants (board)	*pending*
D	Dual encoder, 1 CPU epoch (held-out)	MRR 0.51, top-1 36%

- **Check proxies against the board early:** ours overstated class-3 reach ~4–5×.
- **The points are in classes 1–2:** rank the right *isomer* first.
- **Hunt train/serve mismatches;** select by held-out validation, not ~124 molecules.

5 Next steps (track E)

Compare variants on the board; finish our own pipeline (target ≥ 0.45, fully compliant); add the class-3 engine only where the main list is weak; pick the two finals by validation + board. **Why:** hosts "reserve the right to check all submissions"; if enforced, strong compliant entries may outrank non-reproducible ones.

(−0.000548 Da) that hidden positive-mode m/z values carry: a ~1.6 ppm shift in mass-error features, vs a ~2.5 ppm typical gap between competing formulas (Fig. 2). Two patched variants submitted (pending).